

2024-2025 Spring
MAT124 Final
18/06/2025

1. Let $f(x, y) = 6 - x^2 - y^2$. Find the tangent plane and normal line equations (symmetric or parametric equations) of the graph of f at the point $P(1, 2, 1)$.
2. Find all critical points of the given function $f(x, y) = x^3 + y^3 - 3xy + 2$ and classify them (i.e., determine whether each critical point corresponds to a local maximum, local minimum, or saddle point).
3. Find the maximum and minimum of the function $f(x, y) = 81x^2 + y^2$ subject to the given constraint $4x^2 + y^2 = 9$ using Lagrange multipliers.
4. The equation $z^3 - zy^2 + yx = 3$ defines z implicitly as a function of x and y . It is given that $z = 2$ when $(x, y) = (-3, 1)$. Evaluate $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(-3, 1)$.
5. a. Reverse the order of integration and evaluate the double integral.

$$\int_0^{\sqrt{2}} \int_0^{2-x^2} x e^{x^2} dy dx$$

- b. Evaluate the double integral

$$\iint_R \frac{2xy}{x^2 + y^2} dA$$

where the region $R = \{(x, y) : x \geq 0, y \geq 0, z \geq 0, x^2 + y^2 \leq a^2\}$

6. a. Convert the triple integral into cylindrical coordinates. Do not evaluate the integral.

$$\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz dz dx dy$$

- b. Use a triple integral in spherical coordinates in order to find the volume of the sphere centered at $(0, 0, 0)$ with radius 3.
7. a. Determine whether the vector field $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$ is conservative and find a potential if it is conservative.
- b. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F}(x, y, z) = z\mathbf{i} - y\mathbf{j} + 2x\mathbf{k}$ along the curve $x = t, y = t^2, z = t^3$ from $(0, 0, 0)$ to $(1, 1, 1)$.

1. The equation of the tangent plane at a point on the surface and the normal line to this plane are given, respectively, by

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \quad (1)$$

$$\left. \begin{aligned} x &= x_0 + f_x(x_0, y_0) \cdot t \\ y &= y_0 + f_y(x_0, y_0) \cdot t \\ z &= z_0 - t \end{aligned} \right\} \quad -\infty < t < \infty \quad (2)$$

Calculate the first partial derivatives of f .

$$f_x = -2x, \quad f_y = -2y \quad \implies \quad f_x(1, 2) = -2, \quad f_y(1, 2) = -4$$

Using (1) and (2) determine the equations.

Tangent plane equation: $z - 1 = -2(x - 1) - 4(y - 2) \implies z = -2x - 4y + 11$

Normal line equation: $\left. \begin{aligned} x &= 1 - 2t \\ y &= 2 - 4t \\ z &= 1 - t \end{aligned} \right\} \quad -\infty < t < \infty$

2. To identify the critical points, find where both $f_x = f_y = 0$ or one of the partial derivatives does not exist.

$$f_x = 3x^2 - 3y, \quad f_y = 3y^2 - 3x$$

$$\left. \begin{aligned} f_x = 0 &\implies 3y = 3x^2 \implies y = x^2 \\ f_y = 0 &\implies 3y^2 = 3x \implies x = y^2 \end{aligned} \right\} \quad x = x^4 \rightarrow x(x^3 - 1) = 0 \rightarrow x_1 = 0, \quad x_2 = 1$$

$$x_1 = 0 \implies y_1 = 0, \quad x_2 = 1 \implies y_2 = 1$$

The critical points are $(0, 0)$ and $(1, 1)$. To classify these points, calculate the second partial derivatives and then find the Hessian determinants.

$$f_{xx} = 6x, \quad f_{xy} = f_{yx} = -3, \quad f_{yy} = 6y$$

$$(0, 0) \rightarrow \left\{ \begin{aligned} &f_{xx} = 0, \quad f_{xy} = -3, \quad f_{yy} = 0 \\ &\begin{vmatrix} 0 & -3 \\ -3 & 0 \end{vmatrix} = 0 \cdot 0 - (-3) \cdot (-3) = -9 < 0 \end{aligned} \right.$$

$$(1, 1) \rightarrow \left\{ \begin{aligned} &f_{xx} = 6, \quad f_{xy} = -3, \quad f_{yy} = 6 \\ &\begin{vmatrix} 6 & -3 \\ -3 & 6 \end{vmatrix} = 6 \cdot 6 - (-3) \cdot (-3) = 27 > 0, \quad f_{xx} > 0 \end{aligned} \right.$$

A local minimum occurs at $(1, 1)$ and a saddle point occurs at $(0, 0)$.

3. Let $g(x, y) = 4x^2 + y^2 - 9$ be the constraint. Solve the system of equations below.

$$\left. \begin{array}{l} \nabla f = \lambda \nabla g \\ g(x, y) = 0 \end{array} \right\} \quad \nabla f = \langle 162x, 2y \rangle = \lambda \langle 8x, 2y \rangle = \lambda \nabla g$$

$$2y = 2\lambda y \implies 2y(1 - \lambda) = 0 \implies y = 0 \quad \text{or} \quad \lambda = 1$$

$$162x = 8\lambda x \implies 2x(81 - 4\lambda) = 0 \implies x = 0 \quad \text{or} \quad \lambda = \frac{81}{4}$$

$$y = 0 \implies g(x, 0) = 4x^2 + 0^2 - 9 = 0 \implies x = \pm \frac{3}{2}$$

$$x = 0 \implies g(0, y) = 4 \cdot 0^2 + y^2 = 9 \implies y = \pm 3$$

Evaluate f at the points $(0, 3)$, $(0, -3)$, $(\frac{3}{2}, 0)$, $(-\frac{3}{2}, 0)$ and compare them all.

$$f(0, 3) = 81 \cdot 0^2 + 3^2 = 9, \quad f(0, -3) = 81 \cdot 0^2 + (-3)^2 = 9$$

$$f\left(\frac{3}{2}, 0\right) = 81 \cdot \left(\frac{3}{2}\right)^2 + 0^2 = \frac{729}{4}, \quad f\left(-\frac{3}{2}, 0\right) = 81 \cdot \left(-\frac{3}{2}\right)^2 + 0^2 = \frac{729}{4}$$

The minimum value is 9, the maximum value is $\frac{729}{4}$.

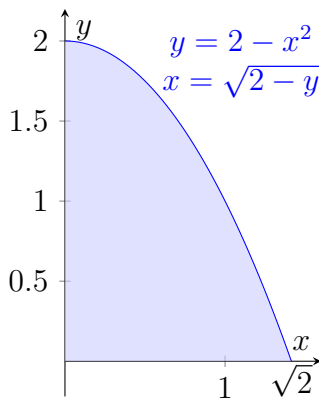
4. Obtain the partial derivatives of both sides.

$$3z^2 \cdot z_x - z_x \cdot y^2 + y = 0 \implies z_x = \frac{-y}{3z^2 - y^2} \Big|_{(-3, 1, 2)} = -\frac{1}{11}$$

$$3z^2 \cdot z_y - z_y \cdot y^2 - 2zy + x = 0 \implies z_y = \frac{2zy - x}{3z^2 - y^2} \Big|_{(-3, 1, 2)} = \frac{7}{11}$$

$\frac{\partial z}{\partial x} = -\frac{1}{11}, \quad \frac{\partial z}{\partial y} = \frac{7}{11}$

5. a. Sketch the region.

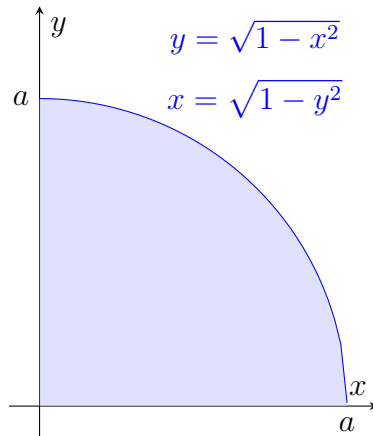


$$I = \int_0^{\sqrt{2}} \int_0^{2-x^2} x e^{x^2} dy dx = \int_0^2 \int_0^{\sqrt{2-y}} x e^{x^2} dx dy$$

$$= \int_0^2 \left[\frac{e^{x^2}}{2} \right]_{x=0}^{x=\sqrt{2-y}} dy = \frac{1}{2} \int_0^2 (e^{2-y} - 1) dy$$

$$= \frac{1}{2} [-e^{2-y} - y]_0^2 = \boxed{\frac{e^2 - 3}{2}}$$

b. Sketch the region.



$$\iint_R \frac{2xy}{x^2 + y^2} dA = \int_0^a \int_0^{\sqrt{1-x^2}} \frac{2xy}{x^2 + y^2} dy dx$$

Notice that it would be difficult to solve the integral in rectangular coordinates. We may switch to polar coordinates to easily evaluate the integral. Use the transformation below.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2, \quad dA = r dr d\theta$$

$$\begin{aligned} I &= \iint_R \frac{2xy}{x^2 + y^2} dA = \int_0^{\frac{\pi}{2}} \int_0^a \frac{2r \cos \theta \cdot r \sin \theta}{(r \cos \theta)^2 + (r \sin \theta)^2} r dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^a \frac{2r^3 \cos \theta \sin \theta}{r^2(\sin^2 \theta + \cos^2 \theta)} dr d\theta = \int_0^{\frac{\pi}{2}} \int_0^a r \sin(2\theta) dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \sin(2\theta) \left[\frac{r^2}{2} \right]_{r=0}^{r=a} d\theta = \frac{a^2}{2} \int_0^{\frac{\pi}{2}} \sin(2\theta) d\theta \\ &= \frac{a^2}{2} \left[\frac{-\cos(2\theta)}{2} \right]_0^{\frac{\pi}{2}} = \frac{a^2}{4} (-\cos \pi + \cos 0) = \boxed{\frac{a^2}{2}} \end{aligned}$$

6. a. Use the transformation below.

$$\begin{aligned} z &= z \\ x &= r \cos \theta \\ y &= r \sin \theta \\ r^2 &= x^2 + y^2 \\ dV &= r dz dr d\theta \end{aligned} \quad \rightarrow \quad \begin{aligned} \sqrt{x^2 + y^2} &= \sqrt{r^2} = r = z_{\text{upper}} \\ x^2 + y^2 &= r^2 = z_{\text{lower}} \\ -1 < y < 1, &< x < \sqrt{1 - y^2} \\ \implies 0 \leq r \leq 1, &-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \\ xyz &= r \cos \theta \cdot r \sin \theta \cdot z \end{aligned}$$

$$\boxed{\int_{-\pi/2}^{\pi/2} \int_0^1 \int_{r^2}^r z r^3 \sin \theta \cos \theta dz dr d\theta}$$

b. We have the sphere $x^2 + y^2 + z^2 = 9$. Use the transformation below.

$$\begin{array}{ll} \begin{array}{l} z = \rho \cos \phi \\ r = \rho \sin \phi \\ x^2 + y^2 + z^2 = \rho^2 \\ dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \end{array} & \rightarrow \begin{array}{l} x^2 + y^2 + z^2 = 9 \implies \rho^2 = 9 \implies \rho = 3 \\ 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq 2\pi \end{array} \end{array}$$

$$\begin{aligned} \text{Volume} &= \int_0^{2\pi} \int_0^{2\pi} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{2\pi} \sin \phi \cdot \left[\frac{\rho^3}{3} \right]_{\rho=0}^{\rho=3} d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{2\pi} 9 \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} \left[-9 \cos \phi \right]_{\phi=0}^{\phi=2\pi} d\theta \\ &= \int_0^{2\pi} (-9 \cos(2\pi) + 9 \cos 0) \, d\theta = \int_0^{2\pi} 18 \, d\theta = 18 \left[\theta \right]_0^{2\pi} = \boxed{36\pi} \end{aligned}$$

7. a. Assume that $\nabla f = \mathbf{F}$ for some potential function f . Then, the mixed partial derivatives of the components must be equal.

$$\frac{\partial F_1}{\partial y} = 1 = \frac{\partial F_2}{\partial x}, \quad \frac{\partial F_1}{\partial z} = 0 = \frac{\partial F_3}{\partial x}, \quad \frac{\partial F_2}{\partial z} = 0 = \frac{\partial F_3}{\partial y}$$

This means $\mathbf{F}$ is conservative on its domain.

$$\frac{\partial f}{\partial x} = y, \quad \frac{\partial f}{\partial y} = x, \quad \frac{\partial f}{\partial z} = z^2$$

$$\int \frac{\partial f}{\partial x} dx = \int y dx = xy + g(y, z) = f(x, y, z)$$

Take partial derivative with respect to y .

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (xy + g(y, z)) = x + g_y(y, z) = x \implies g_y(y, z) = 0$$

$$\int \frac{\partial f}{\partial y} dy = \int x dy = xy + h(z) = f(x, y, z)$$

Take partial derivative with respect to z .

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (xy + h(z)) = h_z(z) = z^2 \implies h_z(z) = z^2$$

$$\int \frac{\partial f}{\partial z} dz = \int z^2 dz = xy + \frac{z^3}{3} + c = f(x, y, z)$$

The potential function for $\mathbf{F}$ is

$$f(x, y, z) = xy + \frac{z^3}{3} + c, \quad c \in \mathbb{R}$$

b. Parametrize the curve and then evaluate the integral.

$$\mathbf{r}(t) = t \mathbf{i} + t^2 \mathbf{j} + t^3 \mathbf{k}, \quad 0 \leq t \leq 1 \implies \mathbf{r}'(t) = dt \mathbf{i} + 2t dt \mathbf{j} + 3t^2 dt \mathbf{k}, \quad 0 \leq t \leq 1$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^1 (t^3 \mathbf{i} - t^2 \mathbf{j} + 2t \mathbf{k}) \cdot (t \mathbf{i} + 2t \mathbf{j} + 3t^2 \mathbf{k}) dt \\ &= \int_0^1 (t^3 - 2t^3 + 6t^3) dt = \int_0^1 5t^3 dt = \left[\frac{5t^4}{4} \right]_0^1 = \boxed{\frac{5}{4}} \end{aligned}$$